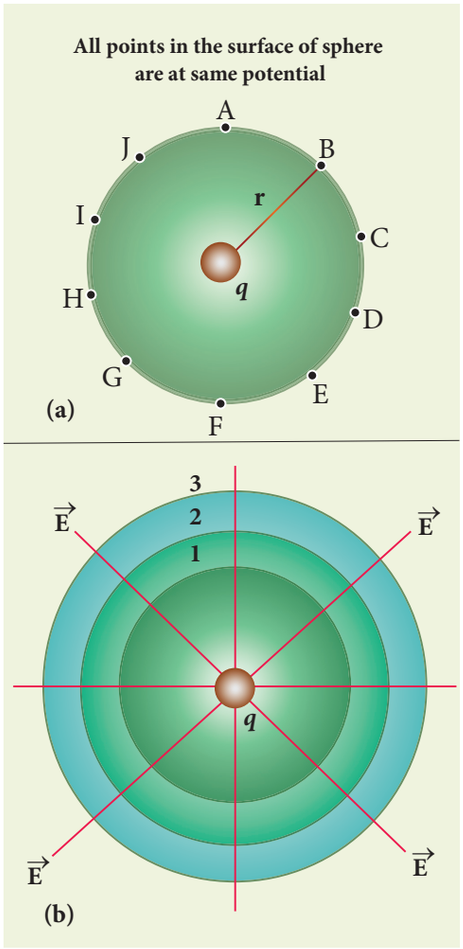



**Figure 1.26** Equipotential surface of point Charge

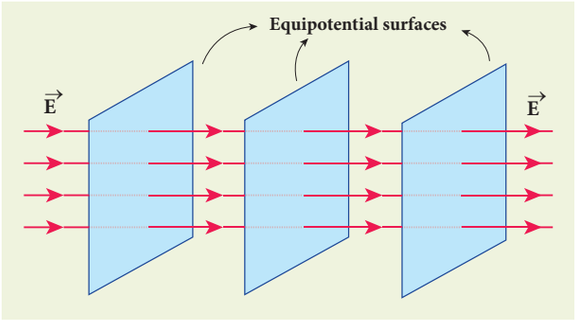



**Figure 1.27** Equipotential surface for uniform electric field

value of the potential is different for different spherical surfaces.

For a uniform electric field, the equipotential surfaces form a set of planes normal to the electric field $\vec{E}$. This is shown in the Figure 1.27.

**Properties of equipotential surfaces**

**(i)** The work done to move a charge q between any two points A and B, $W = q\,(V_B - V_A)$. If the points A and B lie on the same equipotential surface, work done is zero because $V_A = V_B$.

**(ii)** The electric field is normal to an equipotential surface. If it is not normal, then there is a component of the field parallel to the surface. Then work must be done to move a charge between two points on the same surface. This is a contradiction. Therefore the electric field must always be normal to equipotential surface.

## 1.5.5 Relation between electric field and potential

Consider a positive charge $q$ kept fixed at the origin. To move a unit positive charge by a small distance $dx$ towards $q$ in the electric field $E$, the work done is given by $dW = -E\,dx$. The minus sign implies that work is done against the electric field. This work done is equal to electric potential difference. Therefore,

$$dW = dV.$$

$$\text{(or) } dV = -E\,dx \qquad (1.42)$$

$$\text{Hence } E = -\frac{dV}{dx} \qquad (1.43)$$

The electric field is the negative gradient of the electric potential. In vector form,

$$\vec{E} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right) \qquad (1.44)$$

## EXAMPLE 1.14

The following figure represents the electric potential as a function of $x$ – coordinate. Plot the corresponding electric field as a function of $x$.



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